Prema Sai Kuruva

Software Engineer

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Summary

Backend & AI-focused AIML student building FastAPI microservices, ML inference pipelines, and AI-powered developer tools. Experienced with Python, PostgreSQL, FastAPI, Next.js, and LLM integrations.

Experience

SCRS Club - Python Full Stack Developer Aug 2025 - present

- Engineered secure authentication and role-based access control serving 100+ students, managing registration, profile lifecycle, and permission scoping across college-wide deployments.
- Built and integrated REST APIs handling data collection, retrieval, and management workflows across 10+ events and hackathons, supporting 100+ participant records with zero data loss.
- Collaborated within a cross-functional student team to deliver full-stack applications powering institutional activities for a 100+ user base.

Projects

ProthexAI – Clinical Mobility Platform Link
FastAPI, Motor, MongoDB, Google GenAI SDK (Gemini 1.5 Flash), scikit-learn, ReportLab, APScheduler, D3.js

- Designed a dual-layer architecture separating high-frequency sensor streams from hourly aggregates, enabling sub-50ms edge inference while preserving longitudinal clinical context.
- Built a multi-target risk pipeline with independent classifiers per physiological domain, decoupling gait, pressure, and skin-integrity failure modes to prevent cross-axis signal contamination.
- Implemented an AI reasoning bridge grounding probabilistic ML risk scores into plain-language clinical guidance, closing the interpretability gap at point-of-care.
- Automated scheduled PDF report generation with real-time manual-log sync, maintaining consistency between clinical snapshots and live dashboard state.

Praxis – Turn Coding Failures into Learning Intelligence Link
Next.js, React, TypeScript, TailwindCSS, Prisma, PostgreSQL, Groq API, Chrome Extension

- Built a Bayesian inference engine to diagnose root causes of competitive-programming failures directly from submission metadata.
- Designed a PageRank-based graph algorithm to surface recurring weakness patterns and map blind spots in algorithmic knowledge.
- Integrated Groq (LLaMA-3 8B) with a RAG pipeline for personalized learning paths and semantic search across past submissions.
- Developed a Manifest V3 Chrome Extension with TypeScript service workers for real-time LeetCode submission capture and JWT auth.
- Visualized weakness relationships via an interactive React Flow knowledge graph and deployed full-stack on Vercel with sub-second query latency.

Technical Skills

Languages: Python, JavaScript (ES6+), SQL, Java, HTML5, CSS3,TypeScript
Libraries: FastAPI, Next.js, React.js, scikit-learn, TailwindCSS
AI & LLM: RAG, Vector Embeddings, Transformer Models, Prompt Engineering, LangChain, LangGraph
Databases: PostgreSQL, MongoDB, NeonDB, Prisma, MySQL, SQLite 3, Chromadb,Supabase
Tools: Git, GitHub, Docker, Vercel, Render, Postman, VS Code, CI/CD

Education

Kalasalingam Academy of Research and Education (KARE) India
B.Tech in Computer Science and Engineering (AI & ML) *CGPA: 8.62*